Motivation/Main results 4 Minutes

Preliminaries 5 Minutes

Neighbourhood frame

Definition

Let X be a non-empty set and $k \geq 1$. A k-neighborhood frame (k-n-frame) is a pair $(X, \vec{\tau})$, where $\vec{\tau} = (\tau_1, ..., k)$ and each $\tau_i : X \to 2^{2^X}$ assigns to every $x \in X$ a *filter* $\tau_i(x)$ on X, i.e.

- $\tau(x) \neq \emptyset$
- if $U \in \tau(x)$ and $U \subseteq V$ then $V \in \tau(x)$
- if $U, V \in \tau(x)$ then $U \cup V \in \tau(x)$

Full Product of Frames

Definition

Let $\mathcal{X} = (X, \tau_1)$ and $\mathcal{Y} = (Y, \tau_2)$ be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame $\mathcal{X} \times_n \mathcal{Y} = (\mathsf{X} \times \mathsf{Y}, \tau_1', \tau_2', \tau)$ defined as follows :

$$\begin{aligned}
\tau_1'(x, y) &= \{U \subseteq \mathsf{X} \times \mathsf{Y} \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\}, \\
\tau_2'(x, y) &= \{U \subseteq \mathsf{X} \times \mathsf{Y} \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\}, \\
\tau(x, y) &= \{U \subseteq \mathsf{X} \times \mathsf{Y} \mid \exists V \in \tau_1(x)\ \exists W \in \tau_2(y) : V \times W \subseteq U\}
\end{aligned}$$

Bounded morphism

Definition

Let $F = (W, \tau_1, \ldots, \tau_k)$ and $F' = (W', \sigma_1, \ldots, \sigma_k)$ be neighborhood frames. A function $f : W \to W'$ is called a bounded morphism if

- for all $1 \leq i \leq k$, for all $w \in W$ and all $U \in \tau_i(w)$, there is a $V \in \sigma_i(f(w))$ with $V = f(U)$,
- for all $1 \leq i \leq k$, for all $w \in W$ and all $V \in \sigma_i(f(w))$, there exists $U \in \tau_i(w)$ such that $f(U) \subseteq V$.

Note: If $F \twoheadrightarrow F'$, then $Log(\{F\}) \subseteq Log(\{F'\})$.

Definition

Let L_1 and L_2 be two unimodal logics. Then the *fusion* ($\otimes$) of these logics is

$$K_2 + (L_1)_{\Box \to \Box_1} + (L_2)_{\Box \to \Box_2}$$

Furthermore the full product ($\times_n^+$) is defined as

$$L_1 \times_n^+ L_2 = Log\{\mathcal{X} \times \mathcal{Y} \mid \mathcal{X} \vDash L_1 \text{and } \mathcal{Y} \vDash L_2\}$$

Key ideas sketched 15 Minutes

Main Theorem

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \otimes \Lambda \otimes \Lambda + \Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p = \Lambda \times_n^+ \Lambda$$

In the following we will sketch the proof for $\Lambda = T$.

We denote $T \otimes T \otimes T + \Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$ as $T_3 m$

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T^{r,m}_{\omega,\omega,\omega}$) with $Log(T^{r,m}_{\omega,\omega,\omega}) = T_3 m$

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- construct neighbourhood product $N^r_\omega \times^+_n N^r_\omega$ s.t. there is surjective bounded morphism from $N^r_\omega \times^+_n N^r_\omega$ to $N(T^{r,m}_{\omega,\omega,\omega})$

$$T \otimes T \otimes T + (\Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

$\mathbf{Log}(T^{r,m}_{\omega,\omega,\omega}) = T_3m$

- Pick $\mathbb{C} = \{F \mid F \vDash T_3m\}$

Lemma

$Log(\mathbb{C}) = T_3m$

- By Sahlqvist Theorem

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Lemma

$Log(\mathbb{C}) = T_3m$

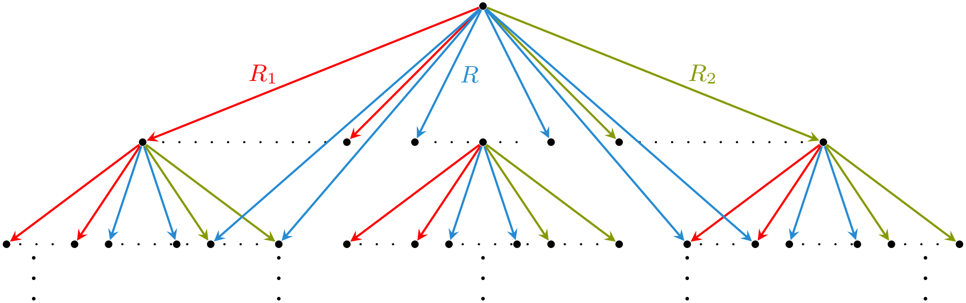
- By Sahlqvist Theorem

Lemma

T_3m has the finite model property

- By Filtration Theorem

$\mathbf{Log}(T^{r,m}_{\omega,\omega,\omega}) = T_3 m$



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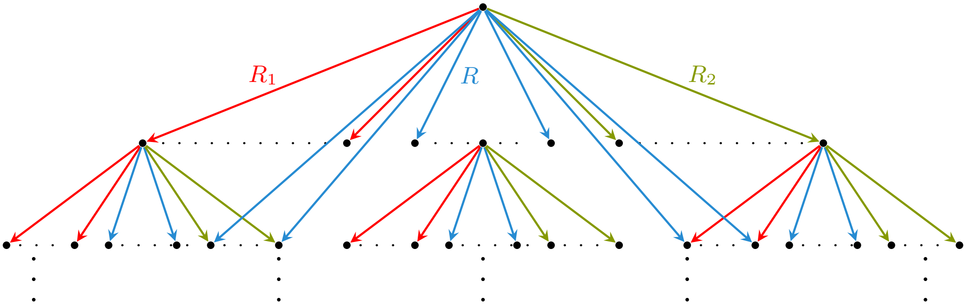

- Note: $\Box p \to \Box_1 p \wedge \Box_2$ is derivable
- Log$(T^{r,m}_{\omega,\omega,\omega}) \subseteq T_3m$ shown by unraveling

Definition of the neighborhood frame N_w^r

Definition

Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0 a_1 a_2 ... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \ \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences.

Definition of the neighborhood frame N_w^r

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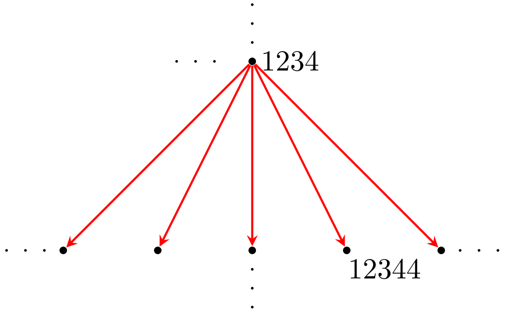
is the set of pseudo infinite sequences. Let $\alpha = a_0 a_1 a_2 ... \in X$. Then

$$U_k(\alpha) = \{\beta \in X \mid \textit{forg}(\alpha) R \textit{forg}(\beta) \text{ and } \alpha \mid_m = \beta \mid_m, \text{with } m = max(k, st(\alpha))\}$$

where *forg*(α) deletes all $\star$ and $st(\alpha) = min\{N \mid \forall k \geq N : a_k = \star\}$

Example of $U_k(\alpha)$



Let $\alpha = \star\star 12 \star 3 \star 4 \star\star^{\omega} \in X$. Then $st(\alpha) = 9$. Let's consider $U_9(\alpha)$. Then

$$\star\star 12 \star 3 \star 4 \star 4 \star\star^{\omega} \in U_9(\alpha)$$

Definition of the neighborhood frame N^r_w

Definition

Let $T^r_\omega = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0a_1a_2... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N}\ \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences. Let $\alpha = a_0a_1a_2... \in X$. Then

$$U_k(\alpha) = \{\beta \in X \mid \textit{forg}(\alpha) R \textit{forg}(\beta) \text{ and } \alpha\mid_m = \beta\mid_m, \text{with } m = max(k, st(\alpha))\}$$

where *forg*(α) deletes all $\star$ and $st(\alpha) = min\{N \mid \forall k \geq N : a_k = \star\}$

We define $N^r_\omega = (X, \tau)$ where $\tau : X \to 2^{2^X}$ with

$$\tau(\alpha) = \{U \subseteq X \mid \exists k \in \mathbb{N} : U_k(\alpha) \subseteq U\}$$

Note: τ is a filter and there is no smallest neighborhood

Theorem

Let $N^r_\omega \times^+_n N^r_\omega = (X \times X, \tau_1, \tau_2, \tau)$ *and* $N(T^{r,m}_{\omega,\omega,\omega}) = (\mathbb{N} \times \{0,1,2\}, \sigma_1, \sigma_2, \sigma)$ *be two full product frames. Then there is a surjective bounded morphism* $g : N^r_\omega \times^+_n N^r_\omega \to N(T^{r,m}_{\omega,\omega,\omega})$. *Note:* $Log(N^r_w) = T$ *and hence* $N^r_w \times^+_n N^r_w \in T \times^+_n T$

Theorem

Let $N^r_\omega \times^+_n N^r_\omega = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T^{r,m}_{\omega,\omega,\omega}) = (\mathbb{N} \times \{0,1,2\}, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N^r_\omega \times^+_n N^r_\omega \to N(T^{r,m}_{\omega,\omega,\omega})$. Note: $Log(N^r_w) = T$ and hence $N^r_w \times^+_n N^r_w \in T \times^+_n T$

Proof sketch: We fix a bijection $h : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$. Let $a = a_0 a_1 ...$ and $b = b_0 b_1 ... \in X$ We define

$$g : X \times X \to (\mathbb{N} \times \{0,1,2\} \cup \{\star\})^* \text{ with } g(a,b) = forg(f(a_0 b_0) f(a_1 b_1) f(a_2 b_2)...)$$

$$f(a,b) = \begin{cases} \star, & \text{if } a = b = \star, \\ (a,1), & \text{if } b = \star, a \in \mathbb{N}, \\ (b,2), & \text{if } a = \star, b \in \mathbb{N}, \\ (h(a,b),0), & \text{otherwise.} \end{cases}$$

Open questions 1 Minute